

Georeference Charts Client Tool

1 Introduction

The Georeference Charts Client is an internal tool used to georeference aviation charts for supported FAA and DOD data sources. The tool supports georeferencing of Instrument Approach Procedures (IAPs) and Airport Procedure Diagrams (APDs) by assigning geographic coordinates to chart reference points.

The Georeference Charts Client allows you to:

- Select and validate chart control points
- Apply georeferencing to procedure charts
- Verify chart alignment using visual reference guides

2 System Requirements

- Windows-based machine required
 - Minimum supported operating system: Windows 10 or later
- Connection to Hilton Software Ethernet (or VPN if working remotely) required
- External monitor strongly recommended for accurate chart scaling. Using only a laptop display may be insufficient to achieve desired visual accuracy

3 Installation

This section includes installation steps.

3.1 Download Application

1. Navigate to the Hilton Software GitLab repository.
2. Open the following repository:
[Georef Charts Client](#)
3. Scroll to the bottom of the repository page and select **Releases**.
4. Download the latest `georef-charts-client.zip` release file.

3.2 Extract Application Files

1. Open File Explorer and locate the downloaded .zip file.
2. Right-click the file and select **Extract All**.
3. Choose the desired extraction location and complete the extraction process.

3.3 Configure Environment Variables

The Georeference Charts Client requires environment variables to be configured before the application is opened for the first time.

1. Open Windows PowerShell.
2. Configure the GeorefServer environment variable based on the intended environment.

3.3.1 Development Environment

Use the Development environment when building or testing locally. The client connects to the ASP.NET development server.

PowerShell
<code>setx GeorefServer "Development"</code>

3.3.2 Production Environment

Use the Production environment when connecting to the live production server.

PowerShell
<code>setx GeorefServer "Production"</code>

3.4 Configure Authentication Token

The Georeference Charts Client requires an authentication token to access the Georeference server when operating on Production environment.

1. Obtain the authentication token from the approved credential management system, which is the folder named Shared-GeorefServer in LastPass.
2. In PowerShell, enter the following command using the provided token value:

PowerShell
<code>setx GeorefSecret "valueFromCredentialManager"</code>

You can close the PowerShell window after all environment variables have been configured. The Georeference client is now installed and ready for initial setup.

NOTE: The required environment variables *must* be configured before launching the application. Launching the tool without the required variables will cause the application to return an exception during startup.

4 Setup

Launch the application.

1. Navigate to the extracted GeorefChartsClient folder.
2. Locate and double-click the application executable (GeoRefCharts.exe) file.

Upon launch, a popup window shows the currently configured environment (i.e. Development or Production).



Figure 1. Environment notification.

Verify that the displayed environment matches the intended server configuration, then select **OK** to continue.

After the environment is confirmed, the Georeference Charts Client displays the login screen, which will require the same credentials the User employs to login to their computer.

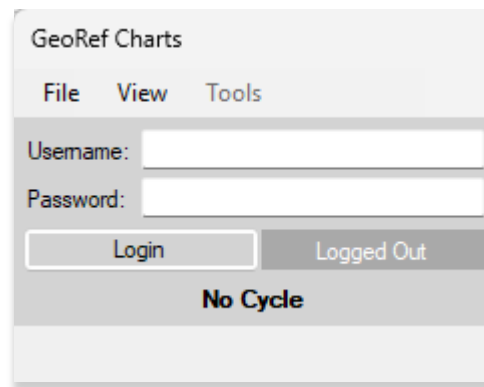


Figure 2. Login screen.

1. Enter your computer username.
2. Enter your computer password.
3. Select **Login**.

If authentication fails, an **Invalid Credentials** message displays. Verify the entered username and password, then attempt to log in again.

After successful authentication, the Cycle Selection window appears. The latest available cycle automatically populates in the cycle field. You may select any of the last three supported chart cycles.

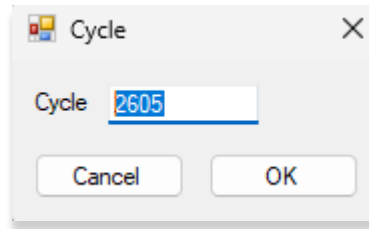


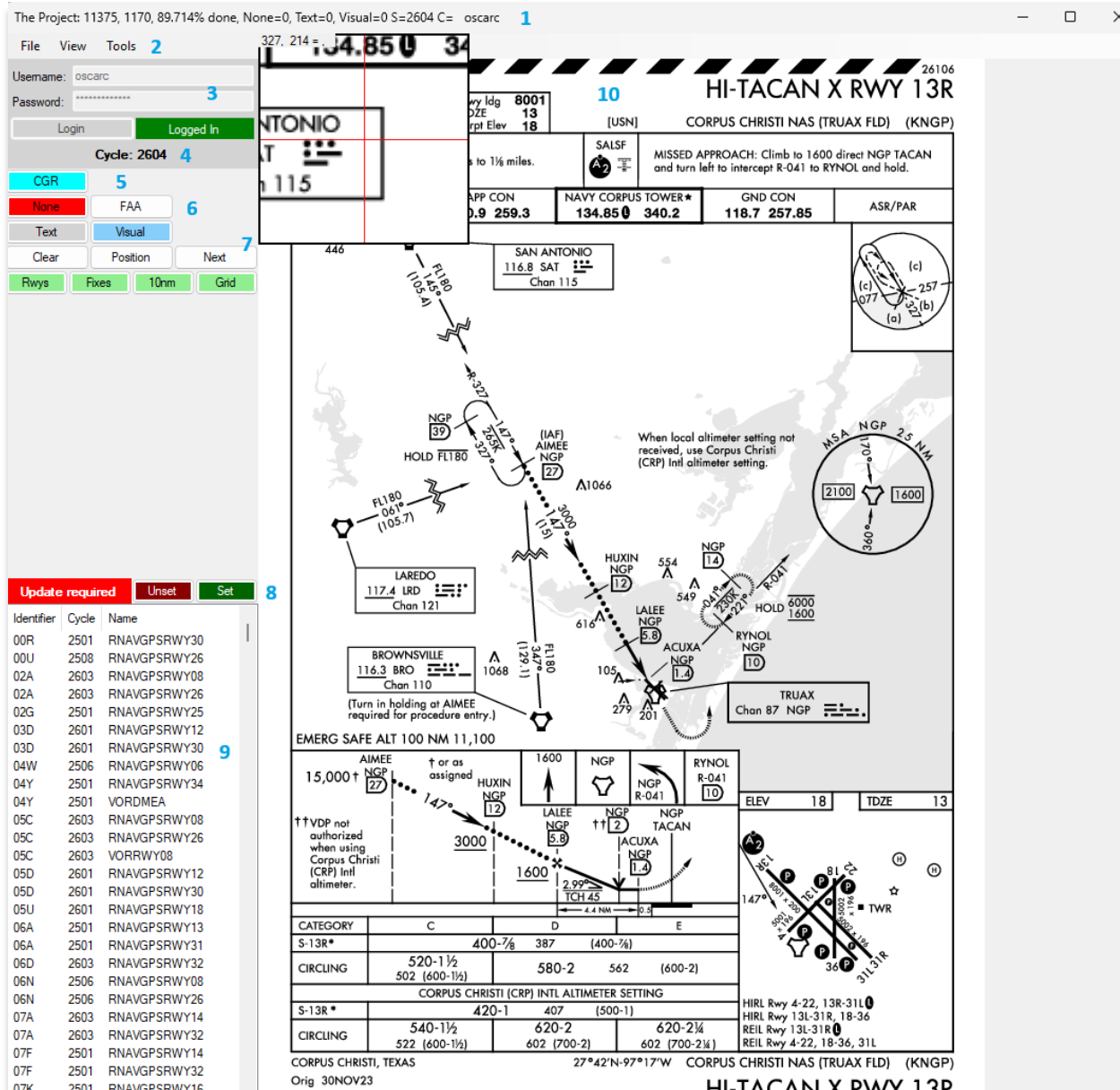
Figure 3. Cycle selection window.

After selecting the desired cycle, continue to the main Georeference Charts Client interface.

5 Tool Overview

After authentication and cycle selection, the Georeference Charts Client main interface displays. The following UI elements can be identified in the picture below by their numeric references:

- Header information bar (1)
- Navigation menu bar (2)
- Credentials and login status (3)
- Cycle switch (4)
- Application modes (5)
- Source modes (6)
- Chart action controls (7)
- Chart status indicators (8)
- Side chart list panel (9)
- Chart view (10)



5.1 Header Information

The header bar is located at the top of the application window and displays chart and user information for the currently selected mode and cycle.

The header includes:

- Total number of charts
- Number of charts pending to be georeferenced
- Georeferencing completion percentage
- Number of charts set to:

- None
- Text
- Visual
- Cycle in which the chart was last modified (S=)
- Cycle in which the chart was last georeferenced (C=)
- Logged-in username

5.2 Navigation Menu

Below the header is the main navigation menu containing the following options:

File

The File menu contains:

- **About** – Displays a brief application summary and version information
- **Exit** – Closes the Georeference Charts Client

View

The View menu controls the application display scale. Available scale values:

- 1.0 (default)
- 1.1
- 1.2
- 1.3

Higher scale values enlarge the application view and chart display. However, changing the application scale on charts georeferenced in previous cycles may temporarily misalign previously selected georeference points. If this happens, switch to another chart and re-select points if the alignment remains incorrect.

Tools

The Tools menu provides chart navigation and chart history functions. The available options include:

- **Go To Airport** – Opens an airport search window where users can search by airport identifier.
- **Go To Last Modified Chart** – Displays the last chart the current user modified.
- **View Last Airport Info** – Displays airport information for the last chart the current user modified.

5.3 Credentials and Login Status

The credentials section displays the current login session information. This section includes:

- Username field
- Password field
- Login status indicator

When successfully authenticated:

- The status button displays Logged In
- The button color changes to green

5.4 Cycle Switch

The Cycle Switch control allows you to change cycles without restarting the application. Selecting the cycle opens the Cycle Selection window, where you can select any of the last three supported chart cycles. After selection, the application reloads the charts for the selected cycle.

5.5 Application Modes

The Georeference Charts Client supports two operational modes.

CGR Mode

CGR (Chart Georef) mode is the default mode and contains all Instrument Approach Procedure charts available for the selected cycle and Source.

APD Mode

APD mode contains all Airport Procedure Diagram charts available for the selected cycle and Source.

Click on the button to switch between modes. The text displayed on the button indicates the active mode.

5.6 Source Modes

The Source Switch allows you to filter charts by chart provider source. The available source types include:

- FAA
- DOD

When a source type is selected, **only** charts contained by said source will appear on the chart list.

Click on the button to switch between modes. The text displayed on the button indicates the active mode.

5.7 Chart Action Controls

The chart action controls provide a variety of operational functions.

None

Used when the chart under analysis can't be georeferenced due to any reason. This action removes the chart from the list and makes it unavailable from the current cycle onwards. The chart will remain unavailable until a source change is made on it. This action should be avoided unless all options have been exhausted.

Text

Used to mark charts that contain only textual information with insufficient chart references for georeferencing. (Available only in IAP mode)

Visual

Used to mark charts containing only visual information without sufficient coordinate references for georeferencing. (Available only in IAP mode)

Clear

Removes all currently displayed georeference points—selected or generated—from the chart view.

Position

Moves the magnifier window to the next corner of the Chart view area in a clockwise order.

Next

Loads a random chart requiring georeferencing from the selected cycle and source type.

Set

Updates the Georeference Server database with the information displayed on the viewed chart (points and their calculated coordinates) and assigns the Current status to said chart.

Unset

Invalidates any previous georeferencing data for the currently viewed chart, making this chart's georeferencing again required.

5.8 Chart Status Indicator

Text box that indicates the loaded chart's status with respect to the currently analyzed cycle. It can contain two different values:

Update Required

Displayed as a red status box with white text. Indicates the chart requires georeferencing updates for the selected cycle.

Current

Displayed as a green status box with white text. Indicates the chart is current and does not require additional georeferencing updates.

5.9 Side Chart List Panel

The side panel displays a list of all available charts for the selected application mode, source type, and cycle. This list contains three columns:

- Identifier (Airport identifier)
- Cycle
- Name

Charts are listed in alphabetical order with respect to Identifier. Selecting a chart from the list loads the chart into the Chart View area.

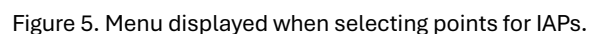
5.10 Chart View

The Chart View is the primary workspace used for georeferencing charts. The selected chart appears in the main viewing area (default scale of 1.0). The Chart View includes:

- Interactive chart display
- Magnifier window
- Real-time coordinate display

The magnifier window appears in one corner of the Chart view. This window is an enlarged, mode detailed view of the area the cursor is hovering on. The coordinates of the exact point the cursor is indicating are displayed above the magnifier window.

To begin selecting a point, user must accurately position the cursor on an identifiable point of the charts, employing the magnifier window for visual clarity and the **keyboard arrows** for fine cursor motion. A right-click on the mouse generates a menu that allows the user to select the type of point they want to link to their current selection. Keyboard shortcuts are also available as shown in the image below.



NavAids

NavAids can typically be identified by their three-character identifiers displayed near the navigation symbol, usually enclosed within a box. Within the box, information such as frequency and Morse code will be provided. Their navigational symbol may appear as a hexagon, compass rose, or circular marker depending on the facility type.

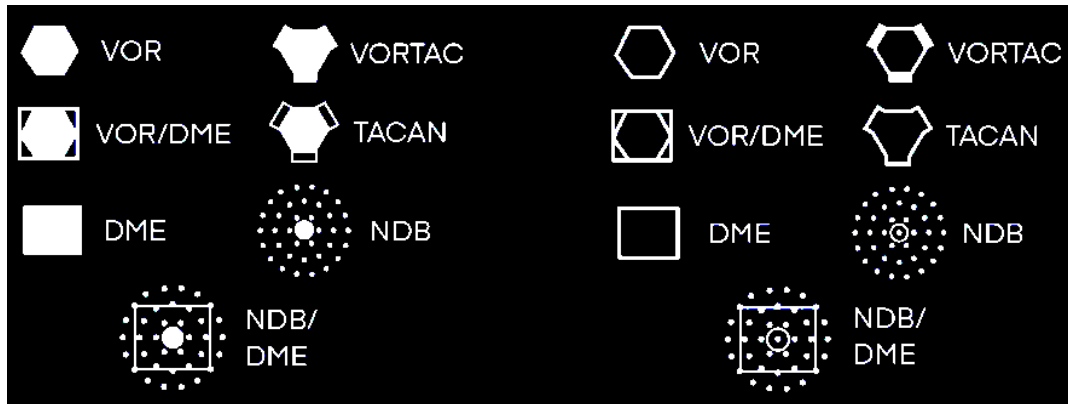


Figure 6. Symbols that represent NavAids.

To choose a NavAid as one of your points, select **NavAid** from the list or enter “=” on your keyboard to open the NavAid selection list. From there, select the correct NavAid identifier.

Waypoint/Fix

Waypoints and fixes can be identified by their five-letter uppercase identifiers (e.g., LNAVY, MAPPs) displayed near a symbol. Their symbols may appear as a small triangle, four-point star, or intersection-style marker depending on the chart type and navigation source.

To choose a Waypoint as one of your points, type the first letter of the desired identifier. The system will display available identifiers for the user to select.

Radial

Radials are determined by their bearing and distance **from** a Navaid. They are labeled with a three-digit magnetic heading and may be shown along the radial line or within a break in the line.

To choose a Radial as one of your points, select **Radial** from the list or enter “[” on your keyboard to open the radial entry view. Enter the associated NavAid identifier, radial value, and distance in the provided fields, then select **OK** to confirm.

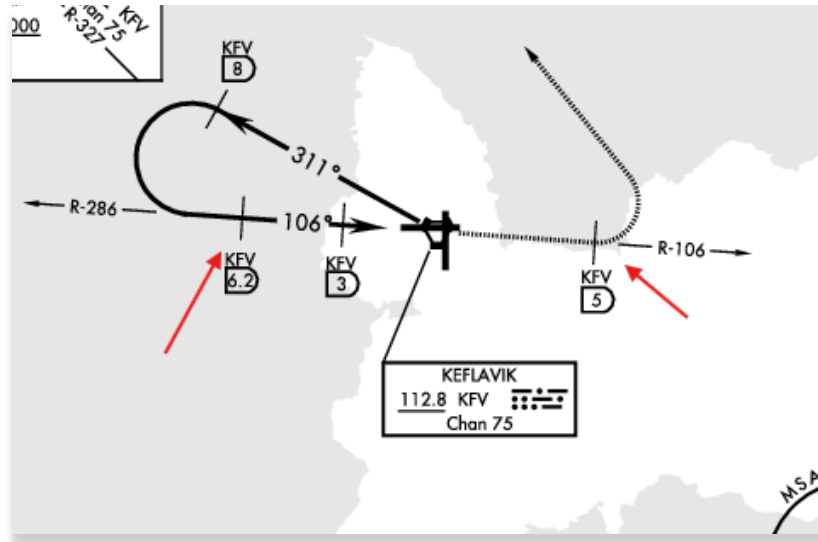


Figure 7. Radials on a chart.

The picture above shows Radials. Taking as examples the elements pointed out by red arrows, we have two radials: [KfV, 286, 6.2] and [KfV, 106, 5]. The **radial** field is the bearing **from** the Navaid, represented by an arrowed line extending away from the Navaid of interest; the **distance** is enclosed in an elongated, D-shaped box.

Runway

Runway threshold points can be prefixed with “RW” followed by a runway number (e.g., RW13, RW34). For parallel runways, an “L” or “R” suffix is included (e.g., RW04L, RW04R). These points may appear as a four-point waypoint-style symbol located at the runway threshold, however, since the runways are visibly identifiable, so are their thresholds even if the star symbol is not present. A detailed view of the runways is usually included in the charts.

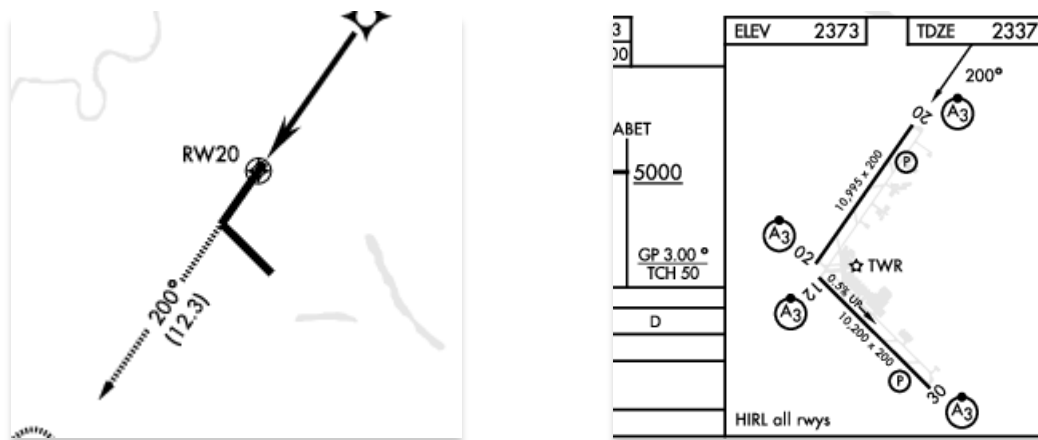


Figure 8. Runways on a chart and their detailed view in the same chart.

To choose a Runway as one of your points, select **Runway** from the list or enter “J” on your keyboard. The runway selection list will appear. Select the matching runway identifier for the selected point.

Obstacle

Although available, Obstacles option should not be used.

Lat/Lon

Entering coordinates manually is an alternative when other chart features can't be accurately located or when the georeference calculated from previously mentioned point types is not satisfactory. This method can be used if the user can obtain external, reliable information (e.g. from Google Maps) about specific geographic points (e.g. rivers, lakes, etc.) and identify those points in the chart.

The user must select Lat/Lon from the menu and enter the latitude and longitude values in the displayed popup.

6.1 Visual Aid Guides

The Georeference Charts Client provides visual alignment guides used to validate the accuracy of georeferencing results and to help identify any issues in the chart alignment, or possible source issues. These guides appear when **all the points have been selected** and are intended for final verification of control point placement. The following visual guides are available:

- **Rwys** – Highlights runway ends. All runway ends are highlighted in a red square and must align precisely with the corresponding runway on the chart.
- **Fixes** – Displays red dots for fixes and waypoints near the airport.
- **10nm** – Draws a 10-nautical mile circumference around the calculated center of the airport. The circle should appear perfectly round when georeferencing is correct. Some charts may also include distance markers that can be used to validate the 10 nautical mile reference.
- **Grid** – Displays reference grid lines with coordinate labels on the chart. This grid is used to validate elements' location.

All these visual aids can be enable/disabled by clicking on the respective buttons in the Chart Action Controls section.

7 Airport Diagram (APD) Georeferencing

Enabled by selecting APD Application mode. Georeferencing an airport diagram requires selecting **three** accurate control points to align the chart correctly with geographic coordinates. The three selected points should form a large triangle and be positioned as far apart from one another as possible.

It is recommended to use runway ends as control points whenever possible, as they are easily identifiable on the chart. If the airport diagram does not contain three runway ends, other identifiable chart features may be used. In these cases, coordinates may be selected from chart references or manually entered into the coordinate fields within the tool.

The process is very similar to that used for CGR mode. After identifying a point on the chart and clicking on it, a menu is displayed. This menu contains three options: Runway, Lat/Lon and Coordlines.

If the **Runways** option is selected, a second menu will be displayed. This new menu will list the available runway ends for the chart. The user must then click on the item that accurately names the selected point.

The **Lat/Lon** option is useful if the exact coordinates for objects in the chart (e.g. buildings, hangars) can be obtained from a reliable source. A right-click on this option will open a popup in which the latitude and longitude for the selected point can be entered.

The **Coordline** option will produce a list of latitudes whose items, in turn, will produce a list of longitudes when the cursor hovers over any of them, linking thus a coordinate pair. This method is convenient when aiming for a point whose coordinates can easily be identified in the chart.

As each point is selected, its coordinates are displayed in the left panel. The first selected point is displayed under **P0**, the second under **P1**, and the third under **P2**.

Once all three points have been selected, a red and blue grid appears on the chart. The red lines in this grid represent geographical meridians, while the blue ones are parallels. Each line has an associated and visible longitude (meridians) or latitude (parallels) for location verification.

Two concentric circles will appear on the runways' ends, blue (inner one) and red (outer one) for those selected by the user, and only red for those calculated by the tool.

Individual control points can be removed if they need to be corrected. A specific point selection can be cleared and re-entered, or all selections may be removed at once using the **Clear** button.